SAND

Recruitment Assignment
Role: Forward Deployed Engineer
Location: Africa

careers.sandtech.com

Forward Deployed Engineer

Recruitment Assignment

The purpose of this assignment is two-fold:

- To give you a better sense of the role of the Forward Deployed Engineer
- To give us a sense of whether you would be a good fit for the role

The assignment is designed to be intentionally diverse and fast-paced, much like the actual role.

As you work through this assignment, note that:

- The structure of the assignment is intentionally broad to give you the freedom to think about the assignment on your own terms. Make sure to include any assumptions you are making as you complete this assignment.
- We understand you don't have enough context on our operations and our business. Working with uncertainty and partial information is a part of this assignment. Be sure to highlight the assumptions you're making and leave comments so we can collaborate with you.
- We understand that in a perfect world, you'd have more time to research and complete this task. Understanding constraints, prioritising, and demonstrating a "done, not perfect" mindset are part of what we look out for. So, in places where you don't have enough time to complete the task to your standards, please share your thought process and your next steps if you have more time.

What you share here is only for assignment purposes. We do not intend to use your submission beyond this recruitment process.

Aside from the submission deadline, the parameters of this assignment are purposefully broad and open to your interpretation and understanding.

We'd like you to return your assignment submission in **one PDF document** on the date stated in the email. Please let us know **at least 24 hours in advance** if you won't be able to submit it on time.

We do not take the evaluation for this role lightly. We appreciate your time, effort and understanding.

We look forward to receiving your assignment!

All the best!

Background

The Health Operating System (HOS) enables multiple data-focused teams to manage and leverage healthcare data effectively, building powerful intelligence to help healthcare administrators optimize service delivery, generate insights, and drive better decisions. As a Forward Deployed Engineer, you will embed with clients to prove value fast, build trust, and deliver tangible ROI through rapid solution engineering.

Assessment Philosophy

This assessment tests your ability in the FDE engagement lifecycle:

- Part 1: Problem Decomposition and Scoping
- Part 2: Rapid Prototyping & Iteration
- Part 3: Optimization & Hardening for Production
- Part 4: Deployment & Knowledge Transfer

We want to see you SCOPE and BUILD. Focus 80% of your time on Parts 1 and 2.

The Scenerio

You are being deployed to one of our expansion countries (i.e. Rwanda) as part of a 2-person FDE team, supported by a solutions manager and country director. The Ministry of Health has signed an MoU with Sand to "improve health data systems and decision-making capabilities." The Solutions manager has identified the following key use cases and solutions opportunities:

- Currently, Maternal and Neonatal Mortality rates are above goals set out by the WHO, Sand Health Operating System (SHOS) can provide immediate value by providing situational awareness
- Currently, there is no single view of the healthcare system, and if interventions are improving the system. Deploying the SHOS Health Atlas provides the ability to visualize and provides situational awareness for improved decision-making.

The following datasets are available

- **DHIS2** - used for monthly reporting (with 2-3 week delays)
- **45 hospitals** - use "HealthTrack" EMR (buggy, local servers)
- **30 clinics** - use OpenMRS
- **175 rural facilities** - paper only
- **Separate systems** - TB program, HIV program (CommCare), immunisation (Excel)
- **Infrastructure** - unreliable power (4-6 hrs/day rural), spotty 3G/4G

Your Internal Support Network:

- Your Country Team: 1 other FDE, 1 Solutions Manager, 1 Launcher/Country Director, 1 Operations Specialist
- FDE Lead: Your performance manager and escalation point
- Central FDEs: Specialist support for technical challenges and pattern documentation
- Product Managers: Own the product roadmap and reusable patterns

Sand Existing Healthcare Products (that you should leverage):

- Health Atlas - Geographic mapping and facility intelligence
- Health Outcome Tracker - Patient outcome analytics
- Health Insight Engine - AI-powered analytics and alerts
- Analytics Template Toolkit - Pre-built reporting templates with Apache Superset
- HealthOS Data Models - Standard healthcare data transformations

The Initial Problem

Week 1 Arrival: You arrive on Monday. The MoH Director says, "Our data is a mess. We cannot make good decisions. We need you to fix it."

After Week 1 Discovery, you identify THREE potential problems:

Problem A: The MoH spends 40 hours/month manually compiling a "Quarterly Health Bulletin" from DHIS2 Excel exports. They want it automated.

The bulletin includes: Top 10 facilities by patient volume, maternal health indicators (ANC visits, deliveries, complications), facility performance scores (reporting completeness, timeliness), trend analysis vs. previous quarters.

Problem B: District health officers cannot see real-time facility status (which are operational, which have stockouts, where disease outbreaks are happening). They make decisions based on 3-week-old data.

Problem C: The TB program and HIV program use separate systems (CommCare). Co-infected patients fall through the cracks because no one has a unified view.

What to Prepare

Deliverable 1: Problem Decomposing and Scoping

Prepare a scoping document that covers:

Section 1: Your Discovery Process

- Who you would talk to in Week 1 and what you would ask
- What patterns would you look for when decomposing "our data is a mess"
- What you would observe directly (not just hear about)

Section 2: Problem Selection

- Which of the 3 problems would you choose to solve in a 6-week sprint (and why)?
- The specific, measurable outcome you will deliver
- What is explicitly OUT OF SCOPE
- Key assumptions and what needs validation in Week 2
- Your success metric
- Fallback plan if the problem is harder than expected

Deliverable 2: Rapid Prototyping

For this part, assume you chose Problem A (Quarterly Health Bulletin automation).

Prepare:

Section 1: Solution Design & Architecture

- Simple architecture diagram showing data flow
- Which Sand products you will use and why
- What you will build custom (if anything) and justification
- Build vs. buy decisions for each component

Section 2: Working Prototype CODE

Required: Build a solution that:

- Reads sample DHIS2-like data (provided data)
- Calculates at least 3 bulletin metrics (e.g., top facilities, maternal indicators, performance scores)
- Outputs results in a usable format (simple HTML, or dashboard)
- Submit working code for the solution

Code Expectations:

- Use whatever stack you are fastest in (Python + Pandas, JavaScript, SQL, etc.)
- Focus on proving the concept, not production quality
- Comment your code to show decision-making

Section 3: Implementation Notes

- What shortcuts did you take to move fast?
- What you would show in Week 3 (what works, what is broken)
- What you learned from building this

Deliverable 3: Production Hardening

List the top 5 things you would fix/add to make your prototype production-ready. For each: (1) What is the concern? (2) How would you address it?

Deliverable 4: Handover & Deployment

- What 3 things would you document for the MoH IT team
- Who needs training and what topics to cover
- Your exit criteria (how you know when to transition out)

Submission Requirements

Submit the following

- Your scoping and design documents
- Working code with sample data
- README with setup/run instructions
- Output demonstrating your solution

What's Next:

Be prepared to present your approach and code in the live interview. You will be asked to walk through your decisions, respond to technical scenarios, and demonstrate your problem-solving process.

Tips for Success:

- Focus 80% of your time on Parts 1 and 2 (scoping and building)
- Your code should work and demonstrate the concept

- Show your thinking process, not just final answers
- Be specific - reference actual Sand products and realistic technical choices

GOOD LUCK!